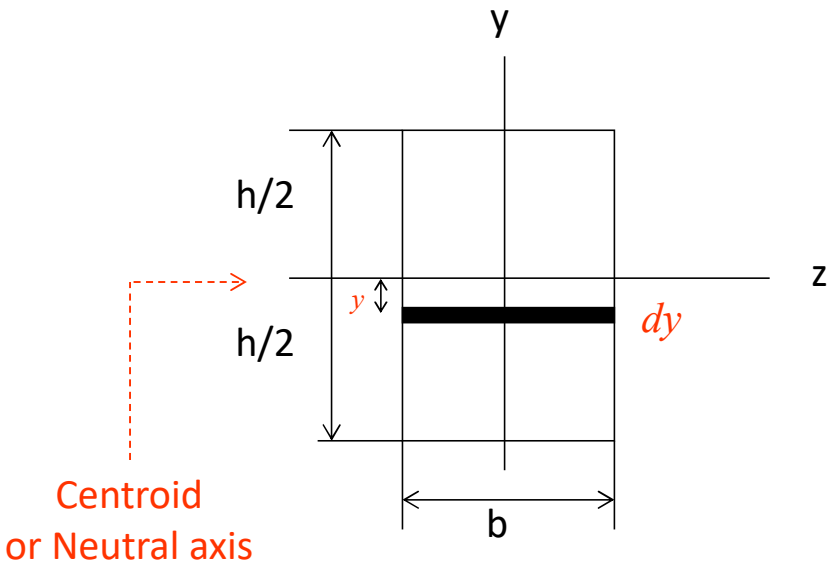
Moment of Inertia
(Second Moment of Area)

Moment of Inertia

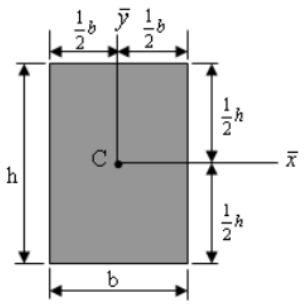
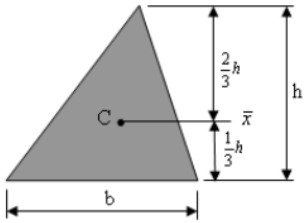
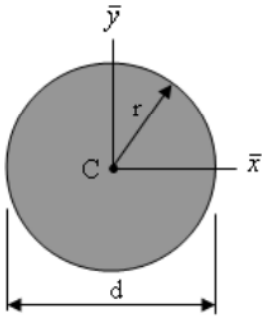
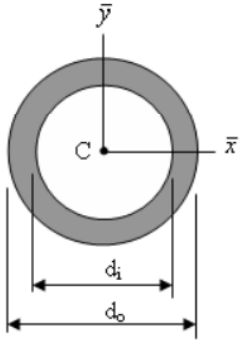
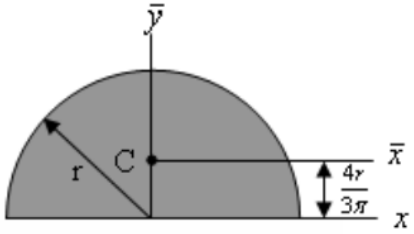
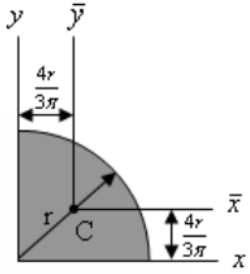
- The moment of inertia is a measure of the amount of resistance a cross-section can offer to bending.
- It is the same as the 2nd Moment of Area of a cross-section and referred to as the Polar moment of inertia in circular cross-sections.
- For the cross-section shown below,



$$I_z = \int y^2 dA = \int_{\frac{-h}{2}}^{\frac{h}{2}} y^2 b dy = \left[b\frac{y^3}{3} \right]_{-\frac{h}{2}}^{\frac{h}{2}} = \frac{b}{3}\left[y^3 \right]_{-\frac{h}{2}}^{\frac{h}{2}} = \frac{b}{3}\left[\frac{h^3}{8} - \left(-\frac{h^3}{8} \right) \right]$$

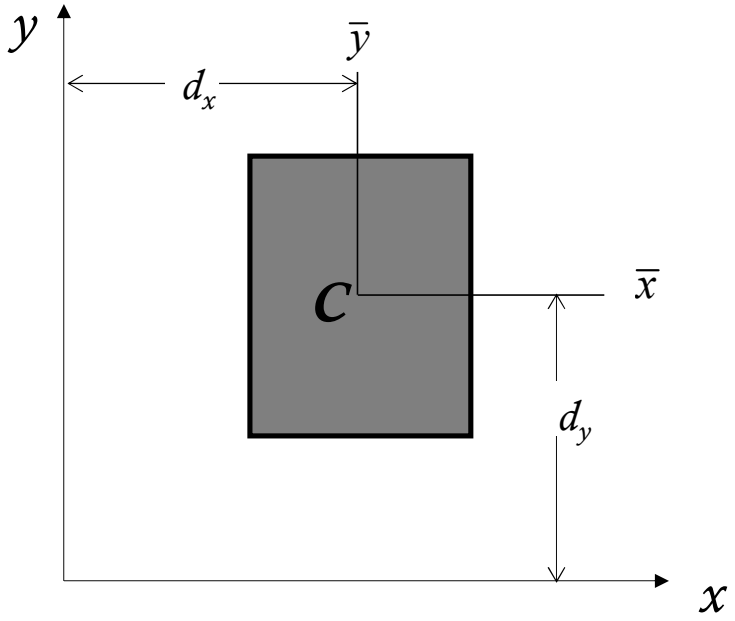
$$\therefore I_z = \frac{bh^3}{12}$$

Moment of Inertia

Shape	Properties	Shape	Properties
	$A = bh$ $I_{\bar{x}} = \frac{1}{12}bh^3$ $I_{\bar{y}} = \frac{1}{12}hb^3$ $r_{\bar{x}} = \frac{1}{\sqrt{12}}h$ $r_{\bar{y}} = \frac{1}{\sqrt{12}}b$		$A = \frac{1}{2}bh$ $I_{\bar{x}} = \frac{1}{36}bh^3$ $r_{\bar{x}} = \frac{1}{\sqrt{18}}h$
	$A = \frac{1}{4}\pi d^2 = \pi r^2$ $I_{\bar{x}} = I_{\bar{y}} = \frac{1}{64}\pi d^4 = \frac{1}{4}\pi r^4$ $r_{\bar{x}} = r_{\bar{y}} = \frac{1}{4}d = \frac{1}{2}r$		$A = \frac{1}{4}\pi\left(d_o^2 - d_i^2\right)$ $I_{\bar{x}} = I_{\bar{y}} = \frac{1}{64}\pi\left(d_o^4 - d_i^4\right)$ $r_{\bar{x}} = r_{\bar{y}} = \frac{1}{4}\sqrt{\left(d_o^2 - d_i^2\right)}$
	$A = \frac{1}{2}\pi r^2$ $I_{\bar{x}} = \left(\frac{\pi}{8} - \frac{8}{9\pi}\right)r^4 = 0.1098r^4$ $I_{\bar{y}} = I_x = \frac{1}{8}\pi r^4$		$A = \frac{1}{4}\pi r^2$ $I_{\bar{x}} = I_{\bar{y}} = \frac{1}{2}\left(\frac{\pi}{8} - \frac{8}{9\pi}\right)r^4 = 0.0549r^4$ $I_x = I_y = \frac{1}{16}\pi r^4$

Parallel Axis Theorem

- Used to determine the *I* of an area about other axes that are parallel to an axis about which the *I* of the area is known (or can easily be determined).



$$I_x = I_{\bar{x}} + Ad_y^2$$

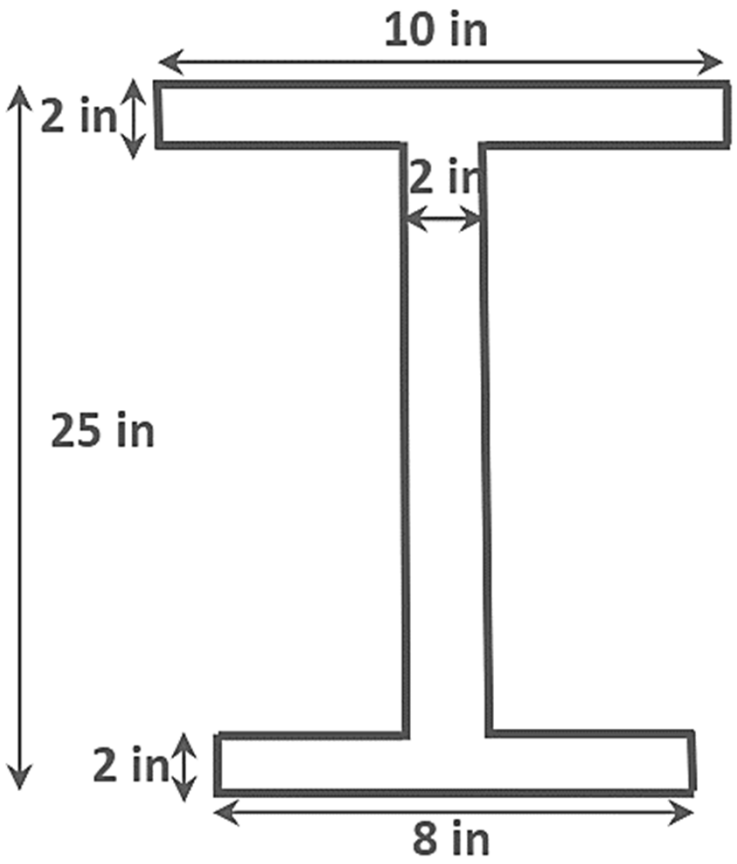
$$I_y = I_{\bar{y}} + Ad_x^2$$

- For composite areas, *I* is be sum of *I* for the "component" areas;

$$I_x = \sum \left(I_{\bar{x}} + Ad_y^2 \right)$$

$$I_y = \sum \left(I_{\bar{y}} + Ad_x^2 \right)$$

Moment of Inertia about the Neutral Axis of a Composite Shape

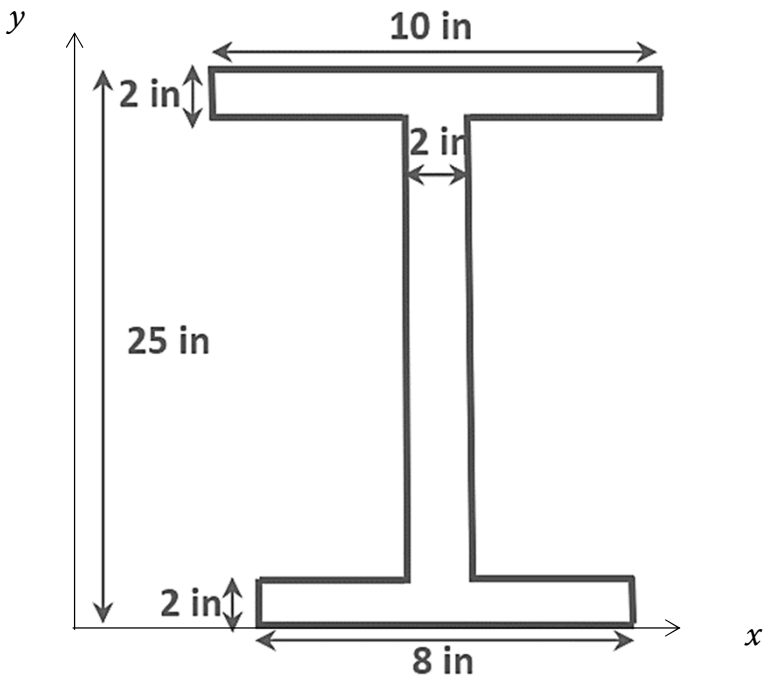


- To determine *I* about the centroidal axis (a line through the centroid) for a composite plane area;
  - Determine the location of the centroidal axis for the composite area.
  - Use parallel-axis theorem to find the *I* of the "component" areas about the neutral axis.
  - Sum the 'component" moments of inertia to get the composite moment of inertia.

- In elastic bending, the neutral axis lies on the axis of symmetry of cross sectional area of the beam, and so coincides with the centroidal axis of the cross sectional area.

Moment of Inertia about the Neutral Axis of a Composite Shape

➢Determine the moment of inertia about the horizontal centroidal axis.



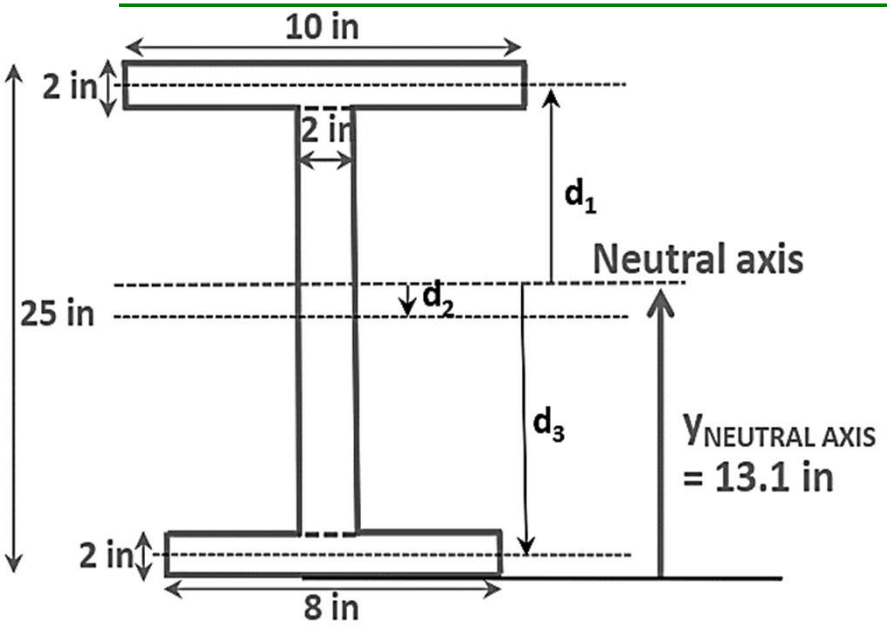
Recall;

$$\bar{Y} = \frac{First\ Total\ Moment\ of\ Area\ about\ x\text{-}axis,\ Q_x}{Total\ Area} = \frac{\sum y_i dA_i}{\sum A_i}$$

Shape	Area (in^2)	$\bar{y}$ (in)	$A\bar{y}$ (in)
Top Rectangle	20	24	480
Middle Rectangle	42	12.5	525
Bottom Rectangle	16	1	16
Σ	78	...	1021

$$\bar{Y} = \frac{1021}{78} = 13.1\ \text{in}$$

Moment of Inertia about the Neutral Axis of a Composite Shape



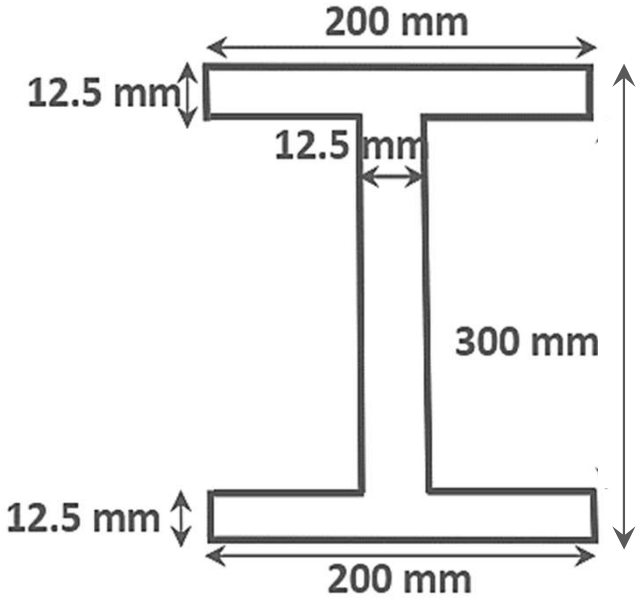
Recall,

$$I = \frac{bh^3}{12}$$

$$I_{N.A} = \sum \left[I_{\bar{x}} + Ad_y^2 \right]$$

Shape	I (in^4)	A (in^2)	d_y (in)	Ad_y^2 (in^4)	$I_{N.A}$ (in^4)
Top Rectangle	6.7	20	10.9	2376.2	2382.9
Middle Rectangle	1543.5	42	0.6	15.1	1558.6
Bottom Rectangle	5.3	16	12.1	2342.6	2347.9
Σ	…	…	…	…	**6289.4**

Moment of Inertia about the Neutral Axis of a Composite Shape



Recall,

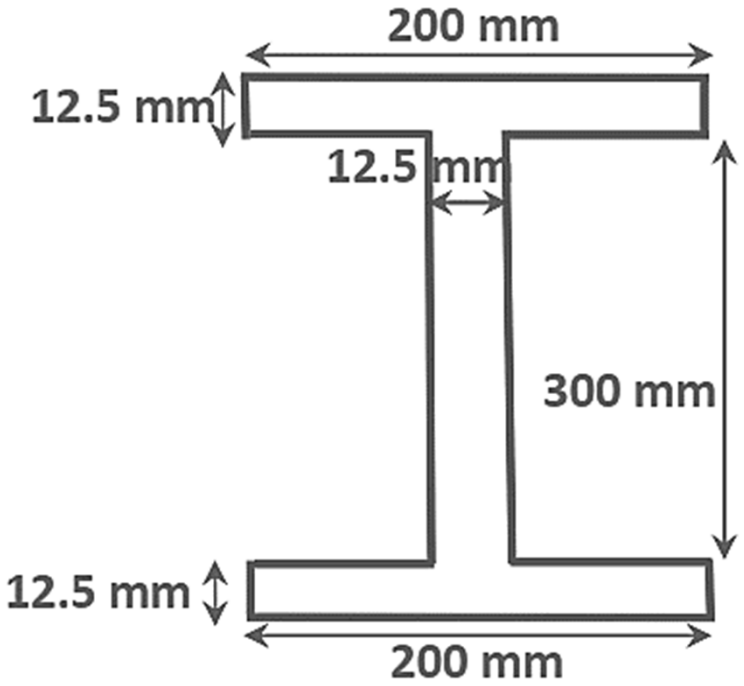
$$I = \frac{bh^3}{12}$$

$$I_{N.A} = \sum \left[I_{\bar{x}} + Ad_y^2 \right]$$

Shape	I (mm^4)	A (mm^2)	d_y (mm)	Ad_y^2 (mm^4)	$I_{N.A}$ (mm^4)
Top Rectangle	32,552.08	2,500	143.75	51,660,156.25	51,692,708.33
Middle Rectangle	21,663,411.458	3,750	0	0	21,663,411.45
Bottom Rectangle	32,552.08	2,500	143.75	51,660,156.25	51,692,708.33
Σ	…	…	…	…	**125,048,828.118**

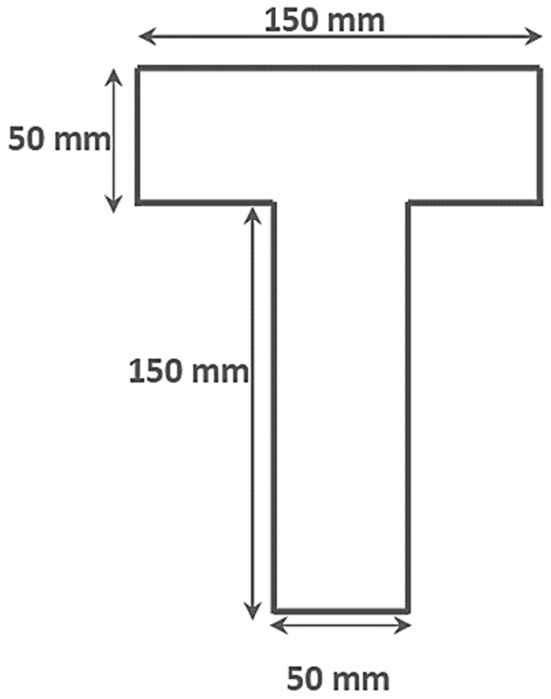
Moment of Inertia about the Neutral Axis of a Composite Shape

- Determine the moment of inertia about the horizontal centroidal axis.



Neutral axis = 162.5 mm from the bottom
I of composite area about neutral axis = 150 x 10^6 mm^4

- Determine the moment of inertia about the horizontal centroidal axis.



Neutral axis = 125 mm from the bottom
I of composite area about neutral axis = 53.125 x 10^6 mm^4